

CHAPTER 25

New challenges in drug discovery

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25.1 Historical and contemporary drug discovery status

Drug discovery is a process by which novel potential medicines are identified via a series of critical research activities (Sang et al., 2018). It comprises a variety of scientific fields, including pharmacology, medicinal chemistry, biochemistry, and biology (Rufer, 2021). The drug discovery centers and industries have supplied several life-saving medications for decades, which contribute to novel treatment opportunities for several life-threatening diseases. A central aim of the initiatives for drug discovery is to identify new molecular entities that can be useful in the management of diseases with unsatisfied needs (Luepke & Mohr, 2017). These conditions do not have useful treatments and, in fact, or potentially, constitute life threats. Many disorders, particularly acute diseases, can now be treated and managed properly (Boon & Edler, 2018). New drugs for heart disease, metabolic disorders, osteoporosis, mental health problems, oncology, gastrointestinal issues, reproductive health care, viral infections, etc. have been discovered to improve health and quality of life (Bano et al., 2019). The entire procedure of drug discovery is a very complicated, time-consuming, expensive, and financially risky effort (Drews, 2000). Nowadays, drugs are produced via different naturally occurring resources, both traditionally or by various innovative techniques (Fig. 25.1). Normally, it takes 10–15 years and requires the application of even over 2.5 billion dollars to have a novel medication in the market (Tichy et al., 2020). In terms of the research & development (R & D) process and the new drug application process (NDA), worldwide the drug development industry is restructuring. Many of the restructurings were driven by costs,



Figure 25.1 Current and emerging sources for drug discovery and development.

often benchmarked by governments in the United States and Europe, to provide the best value to the public purse for the purchase of medicines for national medical services (Downs & Blackburn, 2012). During the early stages of the drug development process, large-scale and complex chemical libraries are examined to identify new hits (i.e., bioactive compounds, ligands, and biomarkers) (Yang et al., 2020). The complete process of drug discovery and development is graphically represented in Fig. 25.2. Traditional medicinal chemistry strategies are often combined with sophisticated structural-based drug design (SBDD) and ligand-based drug design (LBDD) methods for exploring the huge chemical and biological molecules mostly as a major element of hit-to-lead advancement, lead standardization, and development of new active compounds

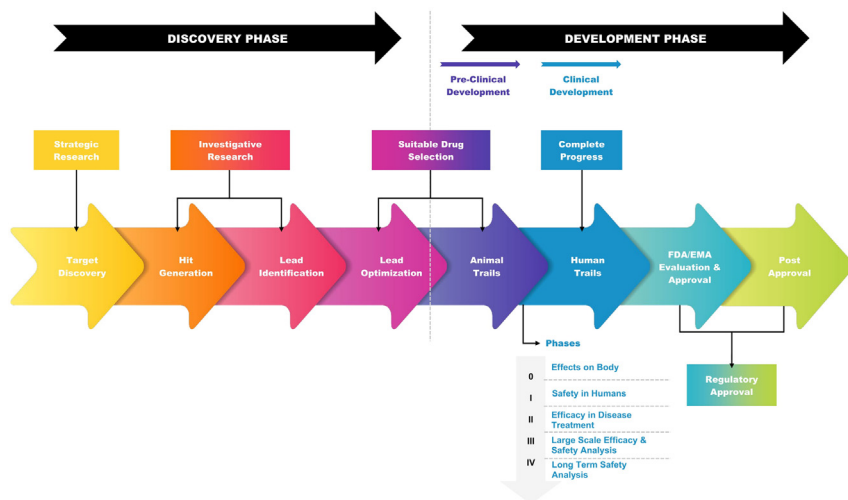


Figure 25.2 Graphical representation of the main and subphases of drug discovery and development process.

(Mozafari, Arab Chamjangali, & Arashi, 2020). It is well known that drug discovery changed. In the past 5 years, Big Pharma around the world has undergone significant restructuring, with large companies remaining in need of all their skills and knowledge. It started outsourcing research organizations but now has gone much even further into the digital world do (Downs & Blackburn, 2012). The path from concept in the new world of drug discovery and the goal of patient benefit validation will be a joint effort (Marshall et al., 2016). It can be demonstrated that new therapeutic classes are generated based on scientific options and therapeutical needs of drugs like pain medication; sedatives, L-dopa, antimicrobial agents, bioactive molecules, immunotherapies, and other drugs. All these medications were developed within a chemical framework of drug therapy and medication (Kardani & Bolhassani, 2021). Historically procedure of drug discovery and development is as old as some of the oldest human cultures. Ayurveda practice in India and traditional Chinese medicines in China are still in practice with over 5000-year-old therapy. Since the time Edward Jenner discovered immunization against smallpox, the beginning of modern medicine is considered (Drahansky, 2011). The field developed gradually until 1928, when Sir Alexander Fleming discovered Penicillin after which the field of medicinal chemistry flourished and it became a complex interdisciplinary platform by the end of the twentieth century,

mainly based on synthetic biological chemistry, which extended to multiple physiological specificities (Bennett & Chung, 2001). Just at beginning of the 21st century, research on drug discovery faced new challenges that transformed the classical concept of drug development that had taken place over half a century in practice. Pharmaceutical firms have therefore settled policies to address these problems and identified many key success features (Zhu, 2020). While there are significant obstacles to marketing new drugs, it is a trend in the past 15 years that the number of drug approvals has increased. The Food and Drug Administration (FDA) authority approved an average of 43 drugs in 2014–18, compared to an average of 23 in the first decade of the century. Researchers regard 2018 as a successful year with almost 59 FDA-approved drugs. Though impressive, these figures must be seen as part of total research and development (R&D) spending in the pharmaceutical sector (Okiyama et al., 2020). As a result, in 35 years, efforts to find and develop new drugs have been doubled. We are still a long way behind this pursuit, despite progress and heavy investment. The concern is that there seems to be something wrong in this translation from precise early discovery experiments to translations in different animal models and clinical trials (Zhu, 2020). All the high molecule assumptions focus on various methods for the mapping of chemicals and wet lab goals, including the target design of drugs, the creation of new targets and implementing existing targets, and the launch of special molecules from multiple perspectives, on a synthetic route and the biodiversity test for novel drugs, appeared to be failed (Du et al., 2020). Despite high promising activities in the in-vitro models, the drop-off of molecules when transported through the development chain is high (Tichy et al., 2020). In recent years, countless emerging innovations and concepts have emerged for drug discovery. Besides the onset of each new method, a commitment was made to generate better drugs while reducing or containing costs (Luepke & Mohr, 2017). At least to some degree, these techniques have been employed in most pharmaceutical companies. During the same period, industry management has been greatly pressured by shareholders and investors to increase drug development and innovation productivity to meet annual growth expectations. The most noticeable methods to cure have already been tried and often failed for many diseases (Gassmann & Reepmeyer, 2005). During the same period, industry management has been greatly pressured by shareholders and investors to increase drug development and innovation productivity to meet annual growth expectations. The most noticeable methods to cure have already

been tried and often failed for many diseases (Cockburn, 2006). For the development and evaluation of new medicines, the complexity of human biology also presents many challenges. Drug molecules with activity specific to one unique biological pathway in diseased cells with an optimal beginning and/or duration of action pharmacokinetic profile or without side effects as a result of systemically exposed conditions are very difficult to design (Silverman et al., 2020). Patient compliance can also be a significant challenge when taking medicines. Drug delivery systems allow the design and engineering level of control to overcome these challenges beyond the inherent properties of the drug molecule (Vargason, Anselmo, & Mitragotri, 2021). The old ways of finding drugs have passed forever and surely nobody wants to go back to packing and running columns manually. The trend we presently see for the large pharmaceutical chemist to become a multidisciplinary drug explorer at the core of a global discovery web (Gassmann & Reepmeyer, 2005). Some common challenges in the field of drug discovery are described in the following sections.

25.2 Common challenges in drug design

The massive number of structural research into biologically necessary proteins shows the general acknowledgment that a possible target structure for the drug company is very valuable knowledge not only for the discovery of lead or for the optimization of leads but also in the later drugs development phases in which problems like toxic effects or bioavailability may arise (Lounnas et al., 2013). Medicinal chemistry is often used as a limiting factor in the development of new medicines because the steps involved in preparing only one compound are labor-intensive. The synthesis and purification bench chemist can only yield a certain series of innovative compounds annually. As mentioned in the above section, the FDA approved 59 new medicinal products in 2018, which is very impressive but only two out of 59 are forecast to accomplish yearly revenues of over 2 billion USD (per annum) when the commercial potential of these drugs is analyzed. In general, average peak sales are projected to decrease for new drugs (Ramezankhani et al., 2020). There may be several reasons for this, but one simple explanation is that many medicines have tiny indications. For example, many new oncology clearances with over 50 new approvals have been available in the last 5 years. Cancer medicines are mostly specific to tissue and tumors and thus only suitable for very small patient subpopulations (Boon & Edler, 2018). However, if additional

subpopulation advantages are demonstrated and additional approvals can be gained, a large commercial potential is still present in cancer drugs. Apart from cancer medicinal products, the newest approvals in the last 5 years represented the so-called orphan medicinal products. Orphan medicines are defined as medicinal products that targeted a small patient population, targeting rare diseases per definition (Gombocz & Vogler, 2020). The consciousness of these illnesses is often short and the marketable potential is frequently not assumed (depending on the patient population). However, authorities give incentives (e.g., tax reduction or exclusive marketing) for the endorsement of orphan medications to foster drug development in orphan diseases (Schulze-Bonhage & Stephani, 2019). Alternative cause for the absence of novel hits is that basic medicines with huge patient populations are already available for many indications. For example, in 2004–10, atorvastatin reported revenues exceeding 10 billion per year. However, revenues have decreased substantially since the termination of the patent life in 2011 (Xu & Wu, 2020). This means that additional lipid decreasing medicines with a statin-like mechanism do not require and do not have commercial potential. Or, in other words, the fruits are low-hanging. The advances in science and technology, combined with the long-term shifts in the world economy, have helped to boost the process of modern drug discovery and development (Okiyama et al., 2020). Several challenges remain, still, and we must compensate our resources for combating the problems at the basic science level and in developing new drugs and treatment approaches. This challenge is going to continue, but we seem to be far behind in overcoming many diseases while trying to improve our knowledge. The search for new knowledge for any given medical situation is always available (Babrow, 2020). This pathway has many unknown facts, so it's always a challenge to find an alternative medicine for an illness. It is difficult to follow the path of the molecule discovery through the development path and it involves time, money, and multiple disciplines. Although efforts are made to speed up the drug development process using innovative technology, current efforts remain ineffective (Kiriiri, Njogu, & Mwangi, 2020). The factors that should be considered during drug discovery are illustrated in Fig. 25.3. Several new strategies, including interactions between academia and industry that promote the conversion of new technologies to products and public funding, have not helped to conquer quite several diseases (Pliakos & Vens, 2020). The efficiency seems to have gone down. The areas within the developmental chain that must be improved or methods

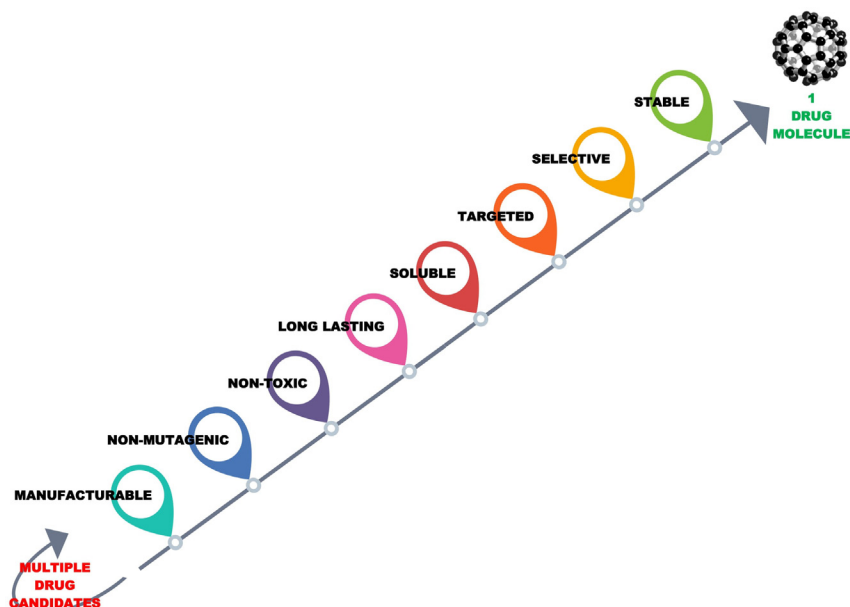


Figure 25.3 Factors that should be considered while selecting a single drug molecule out of many.

used to speed up the process must be identified. All fundamental scientific data have been employed and guidelines developed in therapeutic and pharmaceutical discovery research that have helped to convert the data into useful tools that can be treated or intervene in disease progression (Reker, 2019). Although companies have been finding drugs for more than a century, illustrations of the target molecules are only available in the past decades, most of which are not bound to their cognate ligands. Because the solution of these three-dimensional data has considerably improved, companies have tried to design drugs that deliberately interfere “wise” with the cell function (Gassmann & Reepmeyer, 2005). The advancement of a single therapeutic molecule is by no means the end of the story for infectious diseases. This is often used to produce pathogens, which cleverly developed a method or more for preventing harmful toxicities. This component is used in large measures. This is the case. In the case of cancer, resistance is also an enormous problem. Processes for preventing adverse effects of the medication by conferring resistance organisms and cancer cells are widespread. For example, amino acids in the

target protein can be mutated by the pathogen. Like the HIV mutants, which are nevirapine resistant at the same time. In clinical trials, several potential drugs that result from structural design are well advanced. Considering the number of projects around the world, many more are likely to follow. In a not-too-distant future, several can reach clinics and patients.

25.3 Pharmaceutical Industry challenges for drug discovery

The fundamental strength of pharmacological companies has always been innovation. Again, the drug industry faces several challenges. After the introduction of high-performance synthesis and screening in the early 1990s, the expected increase in the number of strong candidates for development and approved drugs of these technologies was not yet realized (Lee & Kerns, 1999). For the first time in 2001, eighteen new drug entities and seven new biochemical entities were put on the market. It has been the least productive year in the last decade when the majority of additional therapeutic organizations ranged around 44 in 1994 (Showell & Mills, 2003). The pharmaceutical trade faced a significant decrease in productivity, despite large investments. Company size or budget for R&D does not ensure adequate success (Wu & He, 2020). There were almost 4300 players in the pharmaceutical sector over the decades, but only 261 appear to taste success approved by at least one new molecular entity (NME) (Shangary, 2020). At present, only 12% of these companies exist, and the rest are 88% lost or merged with other organizations. Despite significant advances in technological development and considerable R&D investments, the FDA did annual approval of the number of new drug applications in 6 years (2005–2010) with the lowest ratio (20–25 per annum) (Pandey et al., 2020). Remarkably, one of the incipient drifts in 2009 and 2010 was a discrete swing from principal maintenance to specialty goods. Almost 24% (6–5 years, respectively) of drugs approved in 2009 and 2010 belonged to orphan drugs and specialized treatments. Many researchers, conversely, believe that in the next 2–5 years, at best for primary hints the total amount of drug endorsements per annum cannot increase histrionically (Guido, Oliva, & Andricopulo, 2011). Increased drug development costs also complicate this low approval rate. The industry has explored ways to achieve optimal value from the R&D budget with increased drug discovery costs. Both India and China have a vast pool of aptitude, in addition, offer significant savings of 30%–80% of

preclinical and clinical services compared to the costs in the USA. Many major pharmaceutical and international contract research organizations (CROs) have established or are attempting to establish research centers in Singapore, India, and China (Decorte, 2020). These states are booming in competitive assistance required in various aspects of pharmaceutical research and development. Some industries in China and India invest in R&D, albeit predominantly in hazard-shared representations of cooperation. With the declaration of a patent constitution, these industries are going to escalate with the latest R&D, manufacturing, drug testing, and advertising partnerships and business deals (Xu, Wang, & Liu, 2021). Academic groups are increasingly an excellent source of the academic group's new goals and programs, rather than industries considered to have the necessary biological/medical expertise (Verlinde & Hol, 1994).

25.4 Challenges in drug discovery from natural products

The history of medicine goes back to the beginning of human civilization in practical terms. Over the years, modern media cinema or allopathy has progressively developed through the science and experiential efforts of scientists (Patwardhan, Vaidya, & Chorghade, 2004). As always, its development is based on traditional medications and treatments. For centuries, natural products have been a major source of new compound discovery (Bano et al., 2021). Natural products and their derivatives account for nearly 30% of the currently available small molecules (Chen & Kirchmair, 2020). For several years the foundation of traditional medicine has been in use and hence its clinical existence is presumed (Bano & Kašparová, 2021). However, systemic clinical trials are necessary to achieve greater objectivity and confirm traditional claims (Kim, Woo, & Han, 2021). Research into natural chemicals has evolved significantly to include state-of-the-art multidisciplinary approaches that include various methods in drug discovery, molecular genetics, and genomics (De Rycker et al., 2018). Researchers state that new structural types of bioactive natural products play a prominent part in pharmaceutical exploration and development (Atanasov et al., 2021). Furthermore, several techniques (e.g., bioassays for miniaturization, chromato- and spectroscopic coupling, columns of high resolution), have made significant progress in the past few years to explore the vast chemical and biological diversity, enabling the identification and quantification of several bioactive

components (Ahmad Dar, Sangwan, & Kumar, 2020). The identification and selection of valuable lead compounds from several equally active specimens are one of the main challenges in drug discovery (Wilson et al., 2020). In general, once a biologically promising sample has been identified, further bioassays are performed to isolate the active components. To identify the broad range and the intrinsic quality of diverse products leading to new structures and activities, new, more rapid processes are now being developed for the investigation of natural substances using advanced chromatographic techniques as well as online identification of natural productive substances (Atanasov et al., 2021). The biggest hurdles are efficient and faster techniques for the invention and segregation of novel structural bioactive natural products. New techniques and ideologies should therefore be implemented for chemical studies based on natural products for drug discovery (Hon & Lee, 2017). Developments in harvesting, insulation, and purification and analytical and spectroscopic methods have contributed significantly to the quick and reliable elucidation of structural data, enabling the discovery of novel chemical compounds (Matsuda et al., 2020).

25.5 Challenges in SBDD discovery

Historically, long before their biomolecular cognate gene could be segregated and before its chemical structure was established, endogenous bio-compatible ligands were identified. In comparison with much of the information offered by modern experimental chemistry and calculated wave functions, early information regarding the binding site's chemical composition was rudimentary and scarce (Lounnas et al., 2013). This early information, however, often demonstrated the physiological validity of these ligands to the knowledge and development of hypotheses, even if the underlying biological mechanisms were not recognized (Bruno, Miller, & Lim, 2013). Today, the discovery of drugs is powered by biochemical objectives, genetic trials, transgenic livestock models, molecular genetics, gene therapy, and protein science. The configuration of the biomolecular target is therefore of immense potential interest. Drug designers can see ligands attached to their targets utilizing synchrotrons and rapid computers, providing details of unbonded interactions that control the process of binding (Van der Waals disgusting and appealing powers, hydrogens, salt-bridges and water molecules, and ion mediation) (Celikel,

Ruggeri, & Varughese, 2000). Diverse computational techniques have been designed to evaluate binding site interactions, including attempts of estimating the binding affinity of newer molecular entities to their pharmacological activity, including visualizing, docking, molecular simulation, quantum chemistry, gene expression design (Lounnas et al., 2013). For decades, SBDD approaches have been used in bioactive compound design and discovery successfully, with significant improvements in docking protocols and scoring functions, but some challenges still exist in this field (Zhu et al., 2020). Like during the entire modeling phase, target flexibility in molecular docking should be considered. Proteins are displayed as static structures in certain experiments such as X-ray crystallography and SBDD methods. As the structure of protein molecules is indeed constantly developing, and there are more chances of changing the binding site of bioactive pockets leading to protein structure conformations (Bajad et al., 2021). These changes can trigger a changed structure that only affects the biological functions inside chains near a ligand-binding site or subtle change. Ignorance of flexibility in proteins will produce misleading results and reduce the SBDD's accuracy (Degtjarik et al., 2018). Secondly, the organized water molecules are important supporters for many protein-ligand complexes that mediate the interactions between networks of hydrogen connection and influence the free energy of protein-ligand complexes. Besides, the solvation effect is also an important factor in studies of docking for highly solved ligands in particular. However, their contribution is not explicitly estimated because atomic solvation parameters are difficult to transfer to protein-ligand interaction score functions. Further new drug targets and their ligands have been identified with biological technology development (Bajad et al., 2021). Through computational methods or networks, the connection between targets and diseases is thus an efficient way of selecting alternative medicines from large databases. In recent decades, several SBDD software and network services support methods have been used based on various scoring functions (Singh, Tripathi, & Kumar, 2012). These molecular docking programs can predict and estimate the binding and binding free power modes for small molecular ligands within the covalent binding of the receptors. They can then select lead compounds from high and cost ligand databases using experimental methods such as High Throughput Screening (HTS) technology (Rudrapal & Chetia, 2020). In recent years, concerns over the poor prognosis of clinical compounds that have become medicines have concentrated on the quality of compounds to eliminate clinical attrition. Progress

in HTS methods was catalyzed to measure physiochemical characteristics and then developed guiding principles for the design of oral medications by the use of computed properties (Drews, 2000).

25.6 Challenges at preclinical stages of drug discovery

Detailed experimentation in the animal model to analyze whether the drug is suitable for human trials involves the preclinical phase of pharmaceutical research. In particular, it is necessary to monitor and address the toxicity. Moreover, the prospects for the development and safety of drugs require a wide range of strategies in both the preclinical and clinical phases (Lo & Siah, 2021). The two phrases should closely monitor and document potential toxicology related to medicines and the adverse reactions experienced in animal models and patients, including researchers, pharmacologists, clinicians, academicians, health organizations, and industrial stakeholders. There must be an efficient and accessible electronic system to report potential drug issues (Yaqoob et al., 2021). The preclinical stages in drug development have been improved via recent scientific advances, the detection of novel biomarkers, and the upgrading of trials. The lack of knowledge and skills for academic or commercial researchers is an important barrier to drug development (Silvestri et al., 2021). During this stage, animal models that imitate human constraints, like knockouts or transgenic mice, are used. Although a drug is only 12% likely to enter clinical trials in Phase III, it is also expected that the company will produce estimates for an additional drug if it succeeds. Research plans projected via researchers covering an extensive array of chemical, toxicological and biomedical research expertise, that can address not only the fundamental aspects of the scheme but that correspondingly contribute to pharmaceutical growth in preparation and accompanying toxicology and pharmacokinetics revisions in a good laboratory practice (GLP) in the early stage of the development of drugs (Bolon et al., 2020). Approximately 100% of independent scientists' academic research laboratories are not compliant with GLP, therefore data produced in these test centers are not available to the FDA. The magnitude of a research group can determine how much further drug development can be progressed, and these investigating groups are usually not equipped to coordinate research into drug discovery (Sheikh-Hosseini et al., 2020). The principal investigators must publish their ideas or put their ideas in the market of stakeholders who can finance them. Sadly, most scientists attach

themselves too closely to their ideas. The worth of product progress must not only be considered by ideas but also gain to market their concepts to small pharmacological corporations which are specializing in early drug development. Thus, their ideas are allowed to live and flourish in a small pharmaceutical organization that is continuously looking for new prospects and can market and take part in concepts. Once the goal has been proven *in vivo* in animal models and a hit in a chemical series has been discovered, pharmaceutical chemists work. Moreover, for the compound assessment and comparative analysis, the appropriate animal models having virtuous links with human illnesses are critical which is still an obstacle for many illnesses (Shangary, 2020). Therefore, the research models relevant to the exact action mechanism may need to authenticate preclinical mechanisms, but translating such models into progressive human diseases is an independent and at times overwhelming challenge. Animal models show a fairly good translation to human diseases for certain acute indications, for example in antimicrobial therapies. Intended for chronic diseases, however, long-term patterns of disease and suitable secondary species patterns that validate mechanisms with the appropriate biomarkers or the imaging technology would be significant in building sureness and refining the likelihood of victory (Parvatam et al., 2020).

25.7 Challenges at clinical stages of drug discovery

In the field of drug development, clinical trials are very important for the discovery and marketing of new treatments for diseases. The new drug will be tested in 100 or fewer healthy patients during Phase I in clinical trials to compare the significant security of the drug. The proportion of persons improves to a team of 100–500 during Phase II and the effectiveness of the drug is being examined. Such patients have the disease the new medicine is trying to treat. In this stage, side effects, adverse reactions, and effectiveness are all tested. In phase III studies, researchers study a group of approximately 1,000–5,000 patient populations to produce statistically significant data. Only 12% of the drugs pass this phase because it is essential to determine the safety and reliability and effectiveness of the new drug. If a drug has to go through this step, data from a large group of patients is the main foundation for future prescription labeling. The NDA will be forwarded to the FDA for assessment and possible approval, following the success of clinical trials. This document aims to show the safety and effectiveness of the drug and to show that it has been qualified

to go on the market through clinical trials. Many data, including information on all phases of trials, clinical results, precautionary measures, and effects with other medicines are required for that. It may take 6–10 months to conduct the evaluation procedure. Studies can help researchers understand what works and do not work in humans to create the most effective drug possible (Davis et al., 2020). But the challenges of conducting a clinical trial have to be addressed to speed up the process of drug development and concentrate on patient-centered services. The sponsor of the clinical trial is required to offer free medical care for all participant injuries related to the trial. While in public hospitals all participants in a trial can be provided free, and private hospitals are confronted with challenges (Reider, 2015). Therefore, many private hospitals cannot take part in clinical testing. Hence, it is estimated that the budget cuts affected both clinical and preclinical research (Rufer, 2021). As medical research is a long process, hospitals must fund research rather than seek revenue for research dollars. They must have an ethical duty to look beyond profit and practice research as an advertising tool for patient recruitment and health care dollars instead of income for research. We also believe that after a phase I clinical study, a patent clock should start. This minor change could give medicinal industries enormous opportunities to review some of their old projects and the key point should not be the inventiveness of knowledge but also product development should be focused on (Drews, 2000). Other key questions are whether a new trick can be taught on old molecules and the selection of drug candidate molecules for further research. This change to the patent clock will allow investors to plan for comprehensive GLP-compliant toxicology, pharmacokinetics, and efficacy study within a reasonable period and to decide whether to continue in Phase I and Phase II clinical studies (Davis et al., 2020). The latest safety issues relating to experimental and marketing pharmaceuticals should be informed both for medical practitioners and patients. Government education must be provided to residents on the high value of reporting promising drug problems to doctors, drug stores, and health regulators (Castaneda et al., 2015).

25.8 Challenges for small molecule drug design

As therapeutics, peptides, and proteins have great potential, the peptide and protein drug markets are currently estimated to be over 10% of the pharmaceutical market. More than 100 authorized peptide-based

treatments are currently on the market, with a majority less than 20 amino acid residues (Bruno et al., 2013). Small molecules have many advantages, including easily oral administration, cell, and brain permissibility, and cheap operating costs. However, it is not always easy to achieve challenging objectives (Kiriiri et al., 2020). In comparison with the typical small molecule drugs currently significant increase in the number of the pharmaceutical market, proteins, and peptides have multiple contact points with their target (Bruno et al., 2013). In case of difficult or unexplored targets, in particular, new design strategies must succeed. The proteins may not be the only targets on which small molecule compounds can act. Many compounds are considered which bind DNA, but it is always in the area of oncology that medicine kills the cells of the tumor due to quite nonspecific damage to the DNA (Shangary, 2020). The antimalarial agent, for example, quinacrine, forms a complex with genetic material and has anticancer activity. The fairly homogenous structure of the DNA makes it difficult for small molecules to interact (Ramezankhani et al., 2020). RNA is nowadays in the concentration of drug detection. Mostly the inability to find small molecules is because of the difficulty of finding powerful ligands for a latent binding spot. The regional anatomy of the interaction site is critical, particularly for the small molecule protein-protein interactions (PPI) and DNA/RNA-protein boundaries. In several cases, a good safety profile can be determined by the selectivity of closely related homolog compounds. Selectivity can also be a highlight for so-called easily druggable targets. The active site is highly conserved for tyrosine phosphatases, and thus selectiveness, combined with good physiochemical parameters, inhibits drug discovery (Davis et al., 2020). Lately, another way to challenge selectivity has appeared and includes the liquid network underlying the binding spot. Numerous literature revisions show that the water molecule can modulate ligand affinity. This can be cast-off efficiently to modify ligand affinity to diverse proteins selectively and thus acquire selectivity (Celikel et al., 2000). It is rather difficult to predict these effects, but free energy calculations are a promising way forward. The development of small molecular drugs to achieve extremely flexible goals is another hard case (Du et al., 2020). The common key and lock representation of a ligand and goal does not apply for cases in which the ligand encourages the binding pocket outline. In serious situations, inherently distorted protein regions are targeted and ligand binding is ordered. Conventional drug design approaches do not meet these targets and often pharmacological tests detect these molecules (Lavoie et al., 2020).

Strategies to overcome poor bioavailability aim at improving drug solubility through, in particular, altering the local microbial community by applying pH modifiers for small, pH-dependent molecules. This enabled clinical success, such as lactic acid, to improve the solubility of intravenous ciprofloxacin by pH modulation (Vargason et al., 2021).

25.9 Challenges for anticancer drug discovery

The uncontrolled growth of abnormal cells within the body is called cancer. A tumor biomarker refers to the distinctive tumor or the reaction of the body to melanoma that could be analyzed and evaluated objectively (Bhatt et al., 2010). This usually involves assessing genomic modifications or variations in the concentrations of gene expression. Compounds that target genomic changes in the initiation and progression of cancer as a driver of rational therapeutic strategies are widely accepted (Cui et al., 2020). However, only a limited combination of multiple, targeted therapies to treat patients with cancer are still combined with robust, predictive biomarkers. The era of new and innovative biochemical and pharmacological advanced technology has indeed dramatically improved the time from the exploration of the driver's epigenetic modifications to the authorized FDA medication (Haggarty, Karmacharya, & Perlis, 2021). One of the characteristics of tumors is epigenetic instability and the occurrence of aberrations throughout tumorigenesis. Even so, only a small proportion of these changes are true cancer drivers, which have the cellular proliferation benefits of cancer cells; their significant proportion is often better suited for controlling commuters that do not affect the strength and endurance of the malignant clone (Barros et al., 2020). The translation efforts in the clinic to integrate serial tumor samples into patients with different cancers require potential strategies for the development of robust biomarkers despite the challenges of evolving cancer changes (Ileana Dumbrava, Meric-Bernstam, & Yap, 2018). While extensive molecular testing connects the various omics like metabolomics, proteomics, and genomics, may help to identify new biomarkers that can predict response and resistance, a lack of reproducibility of the data between multiplatform and the achieved greatness validation challenges have restricted their regularly scheduled clinical practice implementation (Bensalah, Montorsi, & Shariat, 2007). We can also integrate several biomarkers using computationally demanding analyses and bioinformatics algorithms that have not been constricted to genomic statuses, but it includes transcriptome, methylation,

epigenetic modification, and gene expression (Li et al., 2019). Targeting key pathways and inflammatory cells over the coming decades should include prioritizing the target and treatment based on the limited quality of evidence and developing dynamic clinical combination experiments to diminish the diagnosis and treatment with ineffectual, potentially harmful drugs (Bhatt et al., 2010). Although the current armaments for treatment options including chemotherapeutic drugs, Chemotherapy, and immunotherapy have affected patients, they have posed new challenges in developing and validating prognostic biomarkers for therapies, which will require cooperation and sharing of knowledge between oncologists, immunologists, and molecules (Ileana Dumbrava et al., 2018). Firstly, the novel use of existing medicines is much more economic in comparison with the identification of novel compounds for treating cancer. The second problem is how to handle patient heterogeneity, one of the hardest problems in current cancer therapy. In the last 50 years, many studies have shown that cancer patients have entirely different answers when handled with the same medicine (Hrkach & Langer, 2020). In developing sequence technology, patients can obtain multiomics information before a specific treatment scheme is established. Bioinformatic methods have been used extensively over this decade to interpret sequence data, predict pharmacy (or resistance), and ultimately guide exact drug assortment. Thirdly, multiple medicinal products are commonly used in the treatment of melanoma by combination treatments to improve clinical outcomes. Combination therapies have several benefits over monotherapy. They receive a better capacity to torture cancer cells because a lower dosage can prevent treatment resistance while simultaneously targeting several compensatory paths. They can decrease the risk of complications (Li et al., 2019).

25.10 Challenges in drug discovery via engineered nanoparticles

Engineered nanomaterials are substantially aimed to improve the early detection and specificity of the disease. The limitations of conventional delivery could be overcome from broad-scale issues, such as the distribution of biodiversity to smaller hurdles including intracellular trafficking via cellular services, targeting, specific organelle molecular transport, and other approaches (Singh, 2019). They are designed for either a certain location or for the targeted delivery of therapeutic

drugs at a site. Their formation involves self-containment, which is spontaneously formed by blocks in defensive structures or patterns. Nanoencapsulation offers an opportunity to protect and solubilize fragile compounds, i.e., to deliver compounds that have physical and chemical properties that strongly restrict their oral bioavailability and therefore systemic bioavailability (Mitchell et al., 2021). The US National Science and Technology Council (NSTC) launched in 2000 a National Nanotechnology Initiative (NNI) that describes well-defined initiatives and major challenges in this field to facilitate the implementation and the clinical translation of such successful nanoenabled techniques (Pasut, 2019). These measures have assisted nanotechnology's research and improvement efforts, which are a major part of the research and advancement reported in Nanoparticles (NPs). The NPs can boost the stability and solubility of encapsulated carriers, encourage membrane-cross transport, and increase safety and effectiveness over time. Therefore, NPMs (Nanoparticulate Medications) have been widely used and promising results have been achieved both in vitro and in small animal models. The NPMs' clinical translation is a time-consuming and costly process (Bano et al., 2021). Historically, its development was based on a design model approach, which firstly constructed and physiochemically characterized new delivery systems. Investigation of the connections between biology and engineering, including the understanding of the impact of the pathophysiology of diseases on the aggregation, dispersal, retention, and efficiency of nanomedicine, and the biomedical correlation between the animal delivery system and human in vivo behavior is a key determinant for successful NPMs translation (Hua et al., 2018). There is still so little research to understand fully the correlations in clinical applications between the behavior of NPMs and patient biology, and heterogeneity of illnesses in patients which are the most probable reasons promising NPMs have failed in the clinical trials (Manzar & Mujeeb, 2012). These biochemical challenges can become a huge hindrance to investment in nanomedicine by the pharmaceutical industry. To reduce NPMs investment risk, the pre-clinical data sets require a comprehensive evaluation in animal disease models, relevant to human disease, of therapeutic efficacy, safety, biodistribution, and pharmacokinetics (Patra et al., 2018). Moreover, the benefits of ligand-comparative NPMs in the clinical trials are still negligible, despite accumulation in target areas and treating results

from several preclinical trials (Manzar & Mujeeb, 2012). The optimum launching ligand density on each NPMs surface is still to be calculated and will probably depend on the molecular target's characteristics (e.g., expression, position, internalized oppression rate, and immunogenicity). To translate the formulation into a validated clinical treatment, researchers must, finally, consider minimizing the sophistication of NPMs and consider the end therapeutic effect for human uses (Hua et al., 2018).

25.11 Conclusion and future prospective

The pharmaceutical industry has been led in one of the slightest productive eras in olden times via high letdowns and high rate of declining pipelines. However, the trade remains enthusiastic about tremendous opportunities that present challenges. The sector has a change threshold and is likely to have a better future in the subsequent 10 years. The corporations who accept challenges and retort to them will thrive. Arising models with strategic collaboration will develop themes based on combined wisdom, global accessibility, and expertise. Quality, creativity, and speed are key parameters for success. The trade has a past of high-value generation and will last to overwhelmed challenges with ground-breaking researchers and the intelligent use of technologies. The industry will undoubtedly thrive and develop new milestones. The combination of targets and diseases using computer-based methods or networks is another time-efficient way to gain alternative medicines. Many calculation methods are being used to explore associations of drugs to increase the curative effects. We assume homology design, docking, and medicinal chemistry can be combined successfully to speed up drug discovery. Just as drugs detected at the start of the 20th century have always been a valuable part of reality, but the combination of these technologies is part of the “new deal” we advocate. With evidence-based SBDD, the minds of the scientists are reengaged, and they are proud of their “craft.” With continuous evolution in each field of SBDD, it is credible that the future of discovering more effective drugs is promising with the help of SBDD. The major challenges in disease complexity and the need to bring safer drugs to patients faster and at low costs increasingly face scientists in modern drug developme.

Acronyms used in the chapter.

Acronyms	Full-Forms
CROs	Contract Research Organizations
FDA	Food and Drug Administration
GLP	Good Laboratory Practice
HTS	High Throughput Screening
LBDD	Ligand Based Drug Design
NDA	New Drug Application
NIH	National Institute of Health
NNI	National Nanotechnology Initiative
NPs	Nanoparticles
PPI	Protein-Protein Interactions
R & D	Research and Development
NPMs	Nanoparticulate Medications
SBDD	Structural Based Drug Design

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